

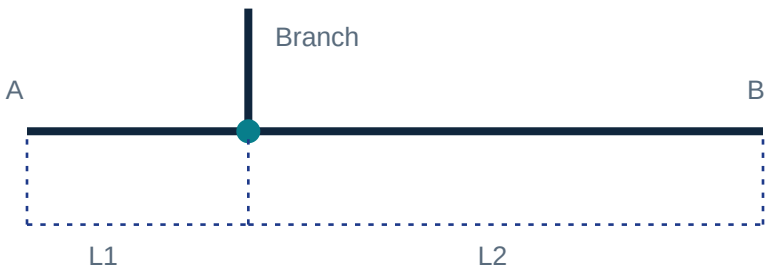
Pipefitter Field Guide

Math, layout and drawing checks for the Pipefitter Field Tool app.

An original pocket guide built around a simple sequence: establish a datum, solve the geometry, account for the actual fittings, then check the finished layout.

Includes worked formulas, rigging calculations, model-specific torch guidance, wraparound marking and folded-paper saddle development. Example dimensions are for training.

Expanded test edition • October 2026. Prepared for hands-on feedback.
Technical review is still pending.



Find it quickly

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Use the right reference

Use this guide to organize measurements and check layout arithmetic. A sketch, a dimension and a fitting catalog each answer a different question. Keep their purposes separate.

The approved drawing defines the intended arrangement. The applicable specification and manufacturer documents identify acceptable components and installation requirements. This guide helps you connect those inputs without silently changing their meaning.

Examples use ideal geometry. Real pipe ends, fittings, joints, supports and access clearances still have to be accounted for. A correct triangle does not prove that a spool will fit.

This test edition is not a welding procedure, pressure-test procedure, engineering design or substitute for site training. Do not infer ratings, torque, wall thickness or allowable loading from its sketches.

FIELD CHECK Before relying on a result, identify the dimension basis, units, source revision and person responsible for approval.

FIELD NOTES / TEST FEEDBACK

Name both ends of a dimension

C-C means center-to-center. E-E means end-to-end. F-F means face-to-face. C-F and C-E connect different reference types. Write down which physical points each letter represents on your particular sketch.

A center may be the theoretical intersection of two pipe centerlines, not a point on the fitting surface. A face may be a flange sealing face; an end may be a prepared pipe end. These references are not interchangeable.

Do not enter a face-to-face measurement into a cut-length calculation expecting a center-to-center value. Convert using the actual component dimensions first.

Leader lines should terminate at the two intended reference points. Put dimension boxes away from crossings so another fitter can trace each value without guessing.



If A is a tee center and B is a flange sealing face, label the span C-F. Record the flange and tee takeoffs separately; do not call that span C-C.

Fractions through sixteenths

Read the whole inches first, then the fraction. The table contains every positive sixteenth below one inch, reduced to its familiar form.

Converting 7/16 to decimal means dividing 7 by 16: 0.4375 in. A decimal such as 0.4375 is part of an inch, not part of a foot.

Fraction	Decimal in	Sixteenths
1/16	0.0625	1/16
1/8	0.1250	2/16
3/16	0.1875	3/16
1/4	0.2500	4/16
5/16	0.3125	5/16
3/8	0.3750	6/16
7/16	0.4375	7/16
1/2	0.5000	8/16
9/16	0.5625	9/16
5/8	0.6250	10/16
11/16	0.6875	11/16
3/4	0.7500	12/16
13/16	0.8125	13/16
7/8	0.8750	14/16
15/16	0.9375	15/16

FIELD CHECK A tape marked in sixteenths cannot justify reporting a measured length to a thousandth of an inch.

Add and subtract cleanly

Put fractions over the same denominator before combining them. Keep the feet and inches visible so a carry or borrow is not lost.

For a long calculation, convert all lengths to inches, work at full precision, then convert the result back to feet and inches. Keep the original field measurement beside the converted value.

A negative answer in a cut-length calculation is a signal to recheck the input references or the available space. It is not a usable pipe length.

Addition: 2 ft 7-3/8 in + 1 ft 8-7/16 in.
Inches: 31-6/16 + 20-7/16 = 51-13/16.
Result: 4 ft 3-13/16 in.

Subtraction: 5 ft 2-1/4 in – 1 ft 9-5/8 in.
62-2/8 – 21-5/8 = 40-5/8 in.
Result: 3 ft 4-5/8 in.

FIELD CHECK Estimate first: about 31 + 20 is about 51 inches. A result near 5 inches or 500 inches needs correction.

FIELD NOTES / TEST FEEDBACK

Round once, at the end

Choose the reporting increment before the final conversion. To round a nonnegative inch value to the nearest 1/16 in, multiply by 16, round to the nearest whole number, then divide by 16.

Keep full precision through intermediate trigonometry. Repeatedly rounding a travel, a takeoff and a remaining span can stack errors.

Rounding is a reporting decision, not a fabrication tolerance. The permitted deviation comes from the job requirements. At an exact halfway point, follow the project rounding convention consistently.

Calculated result: 17.386 in.
 $17.386 \times 16 = 278.176$.
Nearest whole sixteenth: 278.
 $278 \div 16 = 17.375 \text{ in} = 17\text{-}3/8 \text{ in}$.
Rounding changed the value by 0.011 in.

FIELD CHECK Save the unrounded calculation in your notes when a later dimension depends on it.

FIELD NOTES / TEST FEEDBACK

Keep units attached

One inch equals 25.4 millimeters exactly. One foot equals 12 inches or 304.8 millimeters. These are conversion definitions, not estimates. [1]

Convert a mixed feet-and-inches value to inches before multiplying by 25.4.
Convert millimeters to inches by dividing by 25.4.

A unit conversion does not change a product designation into an equivalent purchasable size. Check the specified pipe or tube system rather than selecting a component solely from a converted nominal label.

3 ft 7-1/2 in = 43.5 in.
43.5 × 25.4 = 1104.9 mm.

For 850 mm: 850 ÷ 25.4 = 33.4646... in.
Nearest 1/16 in: 33-7/16 in.

[1] NIST, SI Units – Length.

FIELD NOTES / TEST FEEDBACK

Measure from one datum

Choose a repeatable reference: a marked centerline, an established face or an approved elevation benchmark. Label it A and keep dimensions tied to it.

Record what the tape actually touched. If you measure to the outside of a pipe but need its centerline, document the required geometric correction using the measured or specified outside diameter.

Use an independent overall measurement to check a chain of smaller measurements. Record offsets in separate horizontal and vertical directions before combining them.

Include obstruction envelopes, insulation space and maintenance access on the field sketch when required by the design. A centerline route alone does not describe all the space the assembly occupies.

Datum-to-tee: 24 in. Tee-to-end: 72 in.

Expected overall: 96 in.

If the independent overall is 97 in, reconcile the one-inch difference before cutting.

FIELD CHECK Write the date, drawing revision, instrument and any access limitation beside the survey.

FIELD NOTES / TEST FEEDBACK

Identify the actual component

A nominal pipe-size label is an identifier, not a promise that the measured outside diameter equals the label. Record material, size designation, wall designation and connection type.

For tubing, record the specified outside diameter and wall thickness. Do not substitute a pipe product merely because its description contains a similar number.

Use the exact manufacturer dimensional drawing for the component being installed. Charlotte Pipe, for example, publishes dimensional catalogs for its own pressure pipe and fitting families. [2]

Keep a component schedule: item tag, manufacturer, part number, material, size, end connections, required rating and dimensional source. Leave unknown values visibly unresolved instead of filling them from memory.

FIELD CHECK No pressure/temperature ratings or universal fitting dimensions are reproduced in this guide.

[2] Charlotte Pipe, Pressure Dimensional Catalog.

FIELD NOTES / TEST FEEDBACK

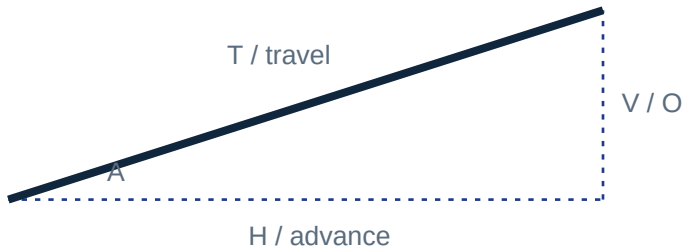
Solve the triangle first

Use a right triangle when two measured components are perpendicular. Name the perpendicular legs H and V, and the diagonal T.

$T = \sqrt{H^2 + V^2}$. If T and H are known, $V = \sqrt{T^2 - H^2}$. A negative quantity inside the square root means the inputs or the assumed geometry do not agree.

For an angle measured from H, $\sin(\text{angle}) = V/T$ and $\tan(\text{angle}) = V/H$. Keep the calculator in degrees when using degree-marked pipe angles. [3]

The triangle describes geometric separation. It does not include elbow center-to-end deductions or joint allowances.



$H = 12$ in and $V = 9$ in.

$T = \sqrt{144 + 81} = 15$ in.

Check: the diagonal is longer than either leg.

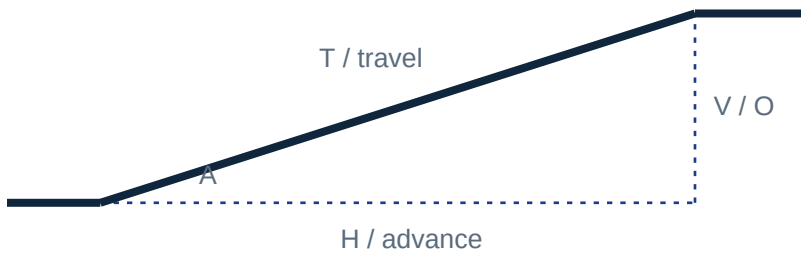
[3] Right-triangle identities; independent example.

Two-bend offset geometry

For parallel entering and leaving centerlines, define O as the perpendicular offset. Let angle A be the direction change of each bend. T is the diagonal distance between the theoretical bend-center intersections, and R is the advance along the original run direction.

From the right triangle: $T = O / \sin(A)$, and $R = O / \tan(A)$. These formulas apply to ideal centerline geometry with equal opposing bend angles.

Choose the bend angle from the approved layout and available components. A small angle needs a longer travel and more run space. Actual fitting bodies and joint requirements must fit within that geometry.



O = 10 in, A = 30°.
 $T = 10 / 0.5 = 20$ in.
 $R = 10 / \tan(30^\circ) = 17.3205$ in.
 The 20 in travel is not the pipe cut length.

FIELD CHECK Before using the formula, confirm the two entering/leaving runs really are parallel.

Offset factors you can check

Multiply perpendicular offset O by the appropriate factor. These values were calculated directly from $1/\sin(A)$ and $1/\tan(A)$; they are not copied fitting dimensions.

The factors are rounded for quick checking. For a final layout, calculate from the angle at full precision, then round the final reported dimension.

Both answers have the same length unit as O. If O is in millimeters, travel and advance are in millimeters.

Bend angle	Travel × O	Advance × O
15°	3.8637	3.7321
22.5°	2.6131	2.4142
30°	2.0000	1.7321
45°	1.4142	1.0000
60°	1.1547	0.5774

For O = 8 in at 22.5°:
Travel ≈ 8 × 2.6131 = 20.9048 in.
Advance ≈ 8 × 2.4142 = 19.3136 in.

FIELD CHECK A 45° offset has advance equal to offset; its diagonal travel is about 1.4142 times the offset.

Worked 45-degree offset

Suppose the approved centerline offset is 18 in and the two direction changes are 45°. Establish the theoretical bend-center intersections on the sketch first.

Calculate travel and advance before selecting a straight pipe cut. Then retrieve the actual takeoff along the travel leg for each fitting.

Joint gaps in this example are invented for arithmetic practice, not a recommendation for any welding process. Real gaps come from the applicable procedure.

Travel = $18 / \sin(45^\circ) = 25.4558$ in.
Advance = 18 in.

Assume verified travel-leg takeoffs of 4 in each, and two specified gaps of 1/8 in:

Cut = $25.4558 - 4 - 4 - 0.125 - 0.125$
Cut = 17.2058 in, or 17-3/16 in to nearest 1/16.

FIELD CHECK Reconstruct the unrounded chain: cut + takeoff A + takeoff B + both gaps must equal travel.

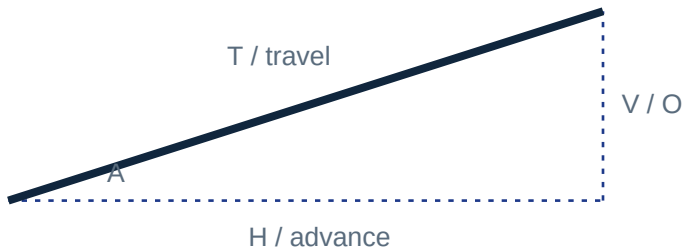
FIELD NOTES / TEST FEEDBACK

A rolling offset uses two triangles

A rolling offset moves sideways and changes elevation. First combine the two perpendicular offsets: true offset $O = \sqrt{\text{side}^2 + \text{rise}^2}$. Then use O in the ordinary two-bend offset formula.

Define the roll angle from the side/horizontal component toward the vertical component: $\text{roll} = \text{atan2}(\text{rise}, \text{side})$. State your viewing direction and sign convention so the assembly cannot be mirrored by accident.

The roll angle orients the offset plane. It is not the elbow bend angle. Those two angles answer different questions.



Side = 8 in, rise = 6 in.

True offset = $\sqrt{64 + 36} = 10$ in.

Roll from the side axis = $\text{atan2}(6, 8) = 36.8699^\circ$.

Using 45° bends: travel = $10 / \sin(45^\circ) = 14.1421$ in; advance = 10 in.

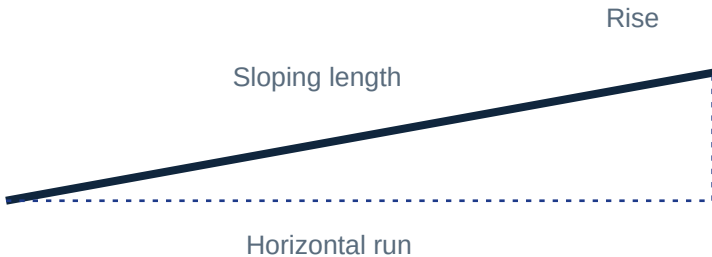
FIELD CHECK Check side and rise independently on the finished layout; the correct diagonal alone can still be in the wrong plane.

Slope is rise over horizontal run

Slope = vertical change / horizontal run. Multiply that ratio by 100 to express percent slope. Keep both values in the same unit.

An instruction stated as inches per foot describes vertical change for each foot of horizontal run. It is not degrees. The sloping pipe length is slightly longer than the horizontal run.

Use only the slope specified for the system. The example below teaches arithmetic and does not prescribe a drainage or process-piping slope.



Specified example: 1/4 in per ft for 16 ft.

Vertical change = $16 \times 1/4 = 4$ in.

Horizontal run = 192 in.

Percent = $4 / 192 \times 100 = 2.0833\%$.

Sloping length = $\sqrt{192^2 + 4^2} = 192.0417$ in.

FIELD CHECK Mark the high and low ends. Do not infer flow direction from the way a drawing happens to face.

Build a dimensional chain

Start with a span whose reference points are explicit. Break it into every physical contribution between those points: fitting takeoffs, the pipe itself, required gaps, and any other specified pieces.

For the simple butt-end example here: C-C span = takeoff A + gap A + pipe cut + gap B + takeoff B. Rearranging gives the pipe cut.

Socket engagement, threads, flange faces and grooved connections use different physical reference points. Draw their chain instead of applying a universal deduction rule.

Dimensions in a catalog may use different datums. Read the sketch and notes attached to the dimension, not just its letter.



C-C = 48 in. Takeoffs = 4 in and 5 in.
 Specified example gaps = 1/8 in at each end.
 Cut = $48 - 4 - 5 - \frac{1}{8} - \frac{1}{8} = 38\text{-}\frac{3}{4}$ in.

FIELD CHECK Kerf is a stock-use allowance; it is not automatically a joint gap in the finished assembly.

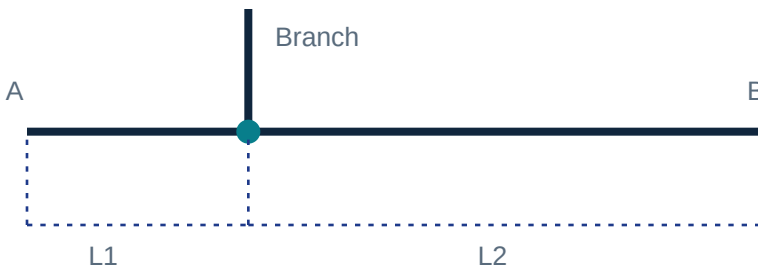
A tee creates separate sections

Give the two run sections and the branch separate identities. A tee center is a junction shared by those sections, not a floating symbol.

If the straight run contains lengths L1 and L2 measured from its end references to the tee center, the tee fraction along that run is $L1 / (L1 + L2)$. With several tees, use cumulative section lengths divided by the run total.

That ratio controls a proportional drawing. Pipe cut lengths still need fitting takeoffs and the correct end references.

When a tee moves on a sketch, inspect every connected branch, flange and nearby dimension. A visually connected line is not proof that all dimensions have been updated.



Run total = 96 in; first section = 24 in.

Tee fraction = $24 / 96 = 0.25$.

The tee belongs one quarter of the way along the run.

For 24, 30 and 42 in sections, tees lie at 25% and 56.25%.

Flanges: face, center and symbol

A flange face is a physical reference surface. A flange symbol is only a drawing convention. Neither its line thickness nor its apparent width on screen provides a fabrication dimension.

In this app, an end flange is drawn as one line across the pipe. The normal flange option is drawn with two lines. Confirm the drawing legend before treating those marks as a bill of materials.

Record the actual flange type, size, material, class, facing and mating details from the specified component documents. Include required face-to-face space and gasket contributions in the correct dimensional chain.

Bolt torque, gasket selection and assembly sequence are joint-specific. Use the approved joint assembly procedure rather than a generic number.

FIELD CHECK If the field measurement reaches the sealing face, label it F. Do not silently move its reference to the flange back or pipe end.

FIELD NOTES / TEST FEEDBACK

Valve layout includes access

Record the valve tag and exact ordered model before using a face-to-face dimension. Different bodies, end connections and actuators can occupy different envelopes.

Show intended orientation and any specified flow arrow. Include operator travel, actuator clearance and maintenance removal space in the layout review.

Draw a dimension chain through the valve assembly using the actual ends or faces. Do not replace the valve with a guessed straight-pipe length in the cutting list.

Keep the functional specification separate from the geometry: a valve that fits between two faces is not necessarily suitable for the fluid, temperature or required service.

If an overall face-to-face envelope is 40 in and a verified valve assembly occupies 12 in of that chain, 28 in remains for the other components and specified joint contributions. That remainder is not automatically one pipe cut.

FIELD CHECK Confirm the valve can be operated and maintained after adjacent pipe is installed.

FIELD NOTES / TEST FEEDBACK

Tube bends: geometry and tooling

For an ideal circular centerline bend, use centerline radius R and bend angle A in degrees. Arc length $B = \pi \times R \times A / 180$. The setback from a theoretical corner to a tangent point is $S = R \times \tan(A/2)$.

For two straight legs meeting at that corner, the ideal developed-length gain is $G = 2S - B$. This is geometric bookkeeping, not a universal bender deduct or arrow setting.

Actual tooling marks and springback corrections depend on the bender, material and tube. Use the manufacturer instructions for the exact tool. [4]

$R = 3 \text{ in}, A = 90^\circ$.

$S = 3 \times \tan(45^\circ) = 3 \text{ in}$.

$B = \pi \times 3 \times 90 / 180 = 4.7124 \text{ in}$.

$G = 6 - 4.7124 = 1.2876 \text{ in}$.

FIELD CHECK Keep CLR, outside radius and inside radius distinct. Do not use a pipe-fitting takeoff as a tubing bend radius.

[4] Swagelok tool guidance; equations and example independently developed.

FIELD NOTES / TEST FEEDBACK

Clock the next bend

A bend angle changes direction within a plane. Clocking rotates that plane around the tube axis before the next bend. Record both values.

Choose a viewing direction, such as looking from the datum end toward the free end. Keep it fixed on the sketch and in the work sequence.

Draw a reference stripe and number the bends in order. Note whether each measurement reaches a tangent, a theoretical center intersection or a finished end.

Before forming the part, check that the chosen sequence can be held in the actual tool without interference. Manufacturer mark conventions take priority over a generic sketch.

Bend record example: B1 = 45°, plane 0°. B2 = 45°, plane rotated 30° clockwise as viewed from datum A. The 30° plane rotation does not replace either 45° bend angle.

FIELD CHECK A mirrored part may have every individual length correct. Verify orientation with the marked datum still visible.

FIELD NOTES / TEST FEEDBACK

Read an ISO as a map

An isometric drawing communicates connections and direction. Unless the drawing explicitly states otherwise, do not measure its printed line lengths with a ruler to obtain fabrication dimensions.

Trace from a known datum or equipment connection. Number each straight section and identify every change of direction, junction, termination and component.

At a crossing, check whether the lines actually join. A tee or connection mark is different from one line passing another.

Keep the north arrow or orientation reference separate from the page edge. Confirm elevations, slope direction and branch orientation with the associated views.

FIELD CHECK Before making a cut list, explain the complete route aloud from one end to the other. Every unexplained jump is a drawing question to resolve.

FIELD NOTES / TEST FEEDBACK

Build the route in the app

In ISO Drawing with PIPE LINE active, tap a start point, release, then tap the next point. Continue with separate taps to add sections.

To branch, tap the body of an existing pipe. The app inserts a tee and splits the run into two sections. The next grid tap starts the branch from that junction.

Tap an existing endpoint to make it the active drawing point. A one-finger swipe pans the view; two-finger pinch changes zoom. The view also follows newly added pipe near the viewport edge.

Drag an endpoint toward an edge to extend the pipe while the view pans. Release to stop. Select END FLANGE for the single-line terminal symbol.

Try a simple run, one tee, one branch and one end flange. Select each section in the list and check that the corresponding pipe highlights.

FIELD CHECK This edition describes the v87 drawing controls; report any phone-specific behavior that differs.

FIELD NOTES / TEST FEEDBACK

Make dimensions easy to trace

Select the intended pipe before editing its measurement. Check whether the value is C-C, E-E, C-F, F-F or C-E, and keep that basis on your field notes.

Drag the dimension box to clear space. Its dark blue dotted lines should follow the box while their pipe ends stay on the section. There is no extra middle connector.

Use DIM – or DIM + for size, FLIP DIM for the opposite side, and ROTATE DIM to change text orientation. Keep a little room between neighboring boxes.

Changing measured straight-run section lengths repositions tees proportionally. Unmeasured sections use inferred drawing lengths until you enter real measurements; replace those estimates before fabrication.

FIELD CHECK Trace both dotted endpoints for every label. A readable number attached to the wrong section is still an error.

FIELD NOTES / TEST FEEDBACK

Prepare a spool record

A useful spool record lets another person rebuild your reasoning. Include the spool identifier, line/service identifier, drawing revision, material and component schedule.

For each section, record its reference-to-reference measurement, both fitting takeoffs, any specified joint contributions, the resulting cut length and the rounding increment.

Keep the raw dimension and final cut in separate columns. Include the source for each non-geometric value, such as a manufacturer part drawing or approved procedure.

Identify match marks and the viewing direction for clocked branches. Record which dimensions are verified, inferred or still awaiting confirmation.

Section	Raw span	Cut	Source/status
S1	C-C 48 in	38-3/4 in	Example only
S2	C-F ____	____	Await face datum
S3	C-C ____	____	Verify branch plane

FIELD CHECK Do not mark a spool ready solely because every cell contains a number.

FIELD NOTES / TEST FEEDBACK

Joining details are not guesses

Check the specified joint type before preparing a pipe end. Welding, threading, socket connections, grooving and mechanical couplings have different dimensional and assembly requirements.

Keep component identification and traceability intact through layout and cutting as required by the project. Record substitutions through the established approval process.

Pressure-test values, hold times, temperature limits, torque values and welding parameters do not appear in this guide. Retrieve them from the controlling system documents and approved procedures.

An app calculation verifies arithmetic only. It cannot determine material compatibility, code acceptance or whether a field assembly is safe to energize.

FIELD CHECK Pause the handoff when component markings, drawing callouts and the material record disagree. Resolve the discrepancy with the responsible person.

FIELD NOTES / TEST FEEDBACK

Plan the work before opening a line

Identify the actual service and hazards before work begins. Stored pressure, thermal energy and unexpected re-energization need the applicable site energy-control process. OSHA describes hazardous-energy controls for covered work; the governing requirements depend on the work and industry. [5]

Hot work also requires control of ignition and exposure hazards. OSHA identifies fire, fumes and radiation among the hazards associated with welding and cutting. Use the applicable permit, supervision and protective measures. [6]

This page is a planning prompt, not an isolation or hot-work procedure. Obtain the site-authorized instructions and trained personnel for the task.

FIELD CHECK Confirm who authorizes the work, which system is involved, how its condition is verified, and which current procedure applies.

[5] OSHA hazardous energy overview. [6] OSHA welding hazards overview.

FIELD NOTES / TEST FEEDBACK

Check the assembly before release

Use independent measurements for the overall dimensions and branch locations. Confirm that the sum of the documented section chains agrees with the overall span.

Review orientation, flange faces, valve access, connection types and any required slope against the approved layout.

Make labels legible on the drawing and on the record. Note revision changes clearly so an older cut list cannot be mistaken for the current one.

Leave unresolved items open. Completion of a drawing or an app checkbox is not acceptance of fabrication, inspection or testing. Record those approvals separately according to the project.

Review item	Record
Overall dimensions	Checked by / date
Tee and branch positions	Datum / result
Faces and orientations	Drawing revision
Component identity	Schedule reference
Inspection or test status	Approved record ID

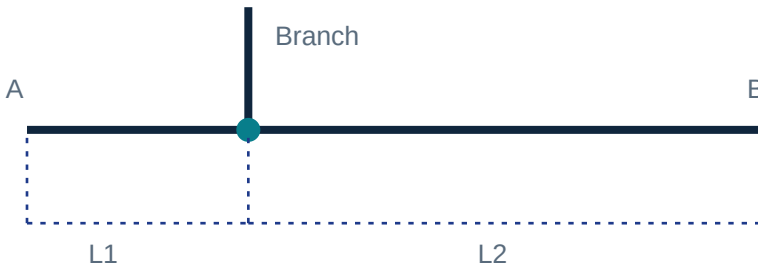
FIELD CHECK Save a copy of the reviewed drawing with its revision and supporting measurements.

Worked spool: two run sections

This practice spool has two straight run spans meeting at a tee. All takeoffs and joint gaps below are hypothetical values supplied for the exercise.

Datum A to tee center = 30 in; tee center to datum B = 66 in. The total is 96 in, so the tee center should appear 31.25% of the way from A to B.

S1 uses takeoffs of 4 in and 3 in with two 1/8 in gaps. S2 uses takeoffs of 3 in and 5 in with two 1/8 in gaps.



$$S1 \text{ cut} = 30 - 4 - 3 - 0.25 = 22.75 \text{ in.}$$

$$S2 \text{ cut} = 66 - 3 - 5 - 0.25 = 57.75 \text{ in.}$$

$$\text{Check S1: } 4 + 0.125 + 22.75 + 0.125 + 3 = 30.$$

$$\text{Check S2: } 3 + 0.125 + 57.75 + 0.125 + 5 = 66.$$

Enter 30 in and 66 in as the run-section C-C values. Do not enter the cut lengths as C-C.

FIELD CHECK The branch needs its own span and takeoff calculation; neither run-section result supplies its cut length.

Working vocabulary

Term	Meaning in this guide
Datum	A stated reference for measurement.
Centerline	The geometric axis of a pipe or tube.
Takeoff	Distance from a stated fitting reference to its connection end or face.
Travel	Diagonal span in the offset triangle.
Advance	Span along the original run direction.
True offset	Combined perpendicular displacement in the offset plane.
Roll	Orientation of an offset plane.
CLR	Radius to the centerline of a circular bend.
Tangent	Where a straight centerline meets a bend arc.
Kerf	Material removed by the cutting process.
Spool	An identified assembly of pipe and components.
Witness line	Line tying a dimension to a reference point.

FIELD CHECK Use the project legend when its terminology differs. State the definition beside an ambiguous measurement.

FIELD NOTES / TEST FEEDBACK

Sources and edition notes

This guide uses original explanations, independently calculated examples and newly drawn diagrams. It does not reproduce Audel text, page layouts, illustrations or tables. Sources below support unit definitions, basic geometry or direct readers to controlling technical information.

Original chapters provide geometry and reference guidance. The practical supplement adds selected model-specific factual settings with attribution; it does not reproduce a complete manufacturer manual. Mention does not imply endorsement.

Edition 0.2 is for app testing and technical review. Before commercial publication, have a qualified technical reviewer check the formulas, examples, terminology and scope. Confirm the final content and illustration rights, and record the reviewer and revision date.

Sources checked October 9, 2026 (UTC).

[1] NIST: SI Units – Length

<https://www.nist.gov/pml/owm/si-units-length>

[2] Charlotte Pipe: Pressure Dimensional Catalog

<https://www.charlottepipe.com/technical-hub/pressure-dimensional-catalog>

[3] OpenStax: Right Triangle Trigonometry

<https://openstax.org/books/precalculus-2e/pages/5-4-right-triangle-trigonometry>

[4] Swagelok: Tube Fitter's Manual resource

<https://www.swagelok.com/en/resources/tube-fitting-manual>

[5] OSHA: Control of Hazardous Energy

<https://www.osha.gov/control-hazardous-energy>

[6] OSHA: Welding, Cutting and Brazing Hazards

<https://www.osha.gov/welding-cutting-brazing/hazards-solutions>

Work a formula, one step at a time

1. Sketch the part. Write the quantity you need and its units. Name every known value on the sketch; do not start by entering numbers into a calculator.
2. Select an equation whose assumptions match that sketch. Rearrange it for the unknown before substituting values. Use parentheses around a whole numerator or denominator.
3. Convert to one unit system. Set trigonometry to degrees for degree angles. Evaluate brackets, powers and roots before multiplication and division. Keep extra digits until the final answer.
4. Put the answer back into the original equation and estimate whether its size makes sense. Show the equation, substitution, result and check in the job record.

Find an offset when travel $T = 20$ in and angle $A = 30^\circ$.

$T = O / \sin(A)$, so $O = T \times \sin(A)$.

$O = 20 \times \sin(30^\circ) = 20 \times 0.5 = 10$ in.

Back-check: $10 / \sin(30^\circ) = 20$ in.

Find A when $O = 10$ in and $T = 20$ in:

$A = \arcsin(O/T) = \arcsin(0.5) = 30^\circ$.

Use inverse sine, not $1/\sin$.

FIELD CHECK An arcsin input outside -1 to +1 signals an invalid ratio for this triangle.

Circumference and equal stations

Use actual outside diameter D for a pattern wrapped around the outside surface. Circumference $C = \pi \times D$. Radius $r = D/2$. A nominal pipe designation is not an outside diameter.

For N equal stations, spacing $s = C/N$ and angular step $= 360^\circ/N$. Measure distances along the paper baseline, not chord distances across the circular end.

Station k sits at $k \times C/N$. Calculate each station from zero rather than repeatedly adding a rounded spacing. The last station closes on the first.

Illustrative actual OD = 4.000 in; N = 16.
C = $\pi \times 4 = 12.56637$ in.
 $s = 12.56637 / 16 = 0.785398$ in.
Angular step = 22.5° .
Station 4 = $C/4 = 3.14159$ in, at 90° .
Station 8 = $C/2 = 6.28319$ in, at 180° .
Station 16 = $C = 12.56637$ in, at 360° .

FIELD CHECK If you measure the circumference directly with paper, use that measured working length for folding. Keep any tape tab outside it.

FIELD NOTES / TEST FEEDBACK

Estimate pipe weight

Prefer documented assembly weights or a verified weighing method. A geometric estimate needs actual dimensions, material weight density and every attached component.

For straight hollow round pipe: inside diameter $d = D - 2t$; metal area $A = \pi \times (D^2 - d^2)/4$; volume $V = A \times L$; weight $W = V \times \gamma$. D , d , t and L must use the same length unit.

Here γ is weight per unit volume, not mass density. If mass is given in kilograms, convert mass to force using $W = m \times g$; at $g = 9.80665 \text{ m/s}^2$, 1000 kg corresponds to 9.80665 kN.

Add fittings, flanges, valves, attachments, contents and other carried items. Include suspended rigging in the hoist load; identify which weights pass through each sling. Unknown contents or weights must be resolved before lifting.

Practice inputs only: $D = 4 \text{ in}$, $t = 0.25 \text{ in}$,
 $L = 120 \text{ in}$, $\gamma = 0.283 \text{ lbf/in}^3$.
 $d = 3.5 \text{ in}$. $A = \pi(16 - 12.25)/4 = 2.94524 \text{ in}^2$.
 $V = 353.429 \text{ in}^3$. $W = 100.020 \text{ lbf}$.
 Add assumed attachments of 40 lbf: 140.020 lbf.
 This estimate is not a verified lift weight.

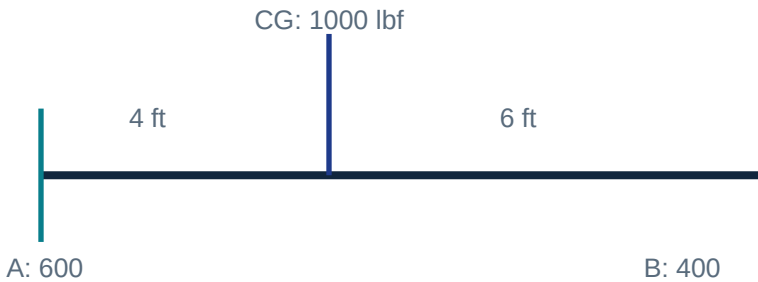
FIELD CHECK A hollow pipe calculation is wrong for a pipe full of liquid, plugged with solids or carrying uncounted fittings.

Find the center of gravity

For known component weights along one axis, $x_{CG} = \sum(W_i \times x_i) / \sum(W_i)$. Choose a common datum and locate each component at its own center of gravity.

For two vertical reactions at A and B, spacing S, with CG a distance x from A: $V_B = W \times x/S$; $V_A = W - V_B$. The support nearer the CG takes more vertical load.

This is a static two-support model. The CG must lie between the supports. A full assembly also needs transverse and vertical CG, stability, attachment strength and clearance evaluated. The hook must be positioned over the combined suspended CG. [8]



600 lbf centered at $x = 2$ ft; 400 lbf at $x = 7$ ft.

Total $W = 1000$ lbf.

$x_{CG} = (600 \times 2 + 400 \times 7)/1000 = 4$ ft.

If A is at 0 ft and B at 10 ft:

$V_B = 1000 \times 4/10 = 400$ lbf.

$V_A = 1000 - 400 = 600$ lbf.

Check moment about A: $400 \times 10 = 1000 \times 4$.

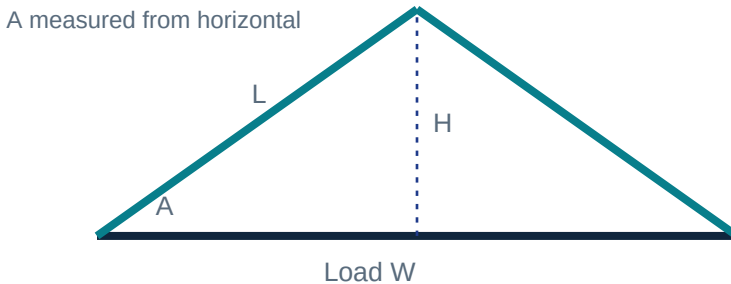
[8] Static equilibrium principles; original example.

Two equal sling legs

For a centered, level load W on two equally loaded straight sling legs, measure angle A from the horizontal load line. Each leg carries vertical share $W/2$. Its tension $T = W / (2 \times \sin(A))$.

If the angle is measured from vertical instead, use cosine. Do not mix the two angle conventions. The included angle at the hook is a third, different angle.

When measuring sling length L from load attachment to hook and vertical rise H , $\sin(A) = H/L$, so $T = (W/2) \times L/H$. These are static forces; shock and acceleration are excluded. [7]



$W = 2000 \text{ lbf}$; $A = 60^\circ$ from horizontal.
 $T = 2000 / (2 \times 0.866025) = 1154.70 \text{ lbf per leg}$.
 At 45° : $T = 1414.21 \text{ lbf per leg}$.
 At 30° : $T = 2000 \text{ lbf per leg}$.
 Check: $2 \times 1154.70 \times \sin(60^\circ) = 2000 \text{ lbf}$.

FIELD CHECK The angles shown are arithmetic examples, not approved configurations. Lift-All advises against angles below 30° from horizontal.

[7] Lift-All angle guidance; examples calculated independently.

Unequal sling loading

Do not divide weight equally when the CG is off center. First calculate each vertical share, then $T_i = V_i / \sin(A_i)$ for the angle of that leg from horizontal.

For a level rigid load with same-elevation attachment points and hook above the CG, let a and b be the horizontal distances from A and B to the hook vertical, and H the rise. Then $LA = \sqrt{a^2 + H^2}$, $LB = \sqrt{b^2 + H^2}$.

$VA = W \times b / (a+b)$; $VB = W \times a / (a+b)$. $TA = VA \times LA / H$; $TB = VB \times LB / H$. The inward horizontal forces must balance in this ideal model. Different elevations or out-of-plane geometry need a different analysis.

W = 3000 lbf; a = 4 ft; b = 6 ft; H = 6 ft.
VA = 1800 lbf; VB = 1200 lbf.
LA = $\sqrt{52}$ = 7.2111 ft; LB = $\sqrt{72}$ = 8.4853 ft.
TA = $1800 \times 7.2111 / 6$ = 2163.33 lbf.
TB = $1200 \times 8.4853 / 6$ = 1697.06 lbf.
Horizontal check: $1800 \times 4 / 6 = 1200 \times 6 / 6$.
Both equal 1200 lbf inward.

FIELD CHECK Neither the average tension nor the smaller tension can be used to select both legs.

FIELD NOTES / TEST FEEDBACK

Turn forces into a lift review

A tension calculation is a demand, not a sling rating. Compare demand with the manufacturer rating for the actual sling, hitch, angle and service conditions. Read whether a tag rates one leg or the complete assembly.

A rated two-leg assembly chart may already include the angle effect. Do not apply the angle factor twice. Do not assume four legs share load equally or that a basket always doubles usable capacity.

Inspect identification and condition before use. Remove defective gear from service; protect slings from damaging edges. Check shackles, hooks, attachment points and the supporting structure as well as the sling. [9]

A qualified lift planner must address load control, CG uncertainty, dynamic effects, hitch stability and any manufacturer reductions. Do not invent a general safety multiplier or use breaking strength as working load limit.

Calculated leg demand = 2163.33 lbf.

An assumed 2000 lbf allowable leg capacity is insufficient.

An assumed 3000 lbf capacity passes only this numerical comparison; the actual configuration, hardware and lift plan still require verification.

FIELD CHECK Keep people clear of suspended loads. Stop if weight, CG, attachment rating or sling identification is unknown.

[9] OSHA 1926.251 and manufacturer rating instructions; this is a training calculation, not a lift plan.

Oxy-acetylene: prepare the setup

This section is for trained operators cutting suitable ferrous material with oxygen and acetylene. Welding, heating, propane and other tip families require their own instructions. Identify the torch, attachment, tip, regulators, hose and protective devices before selecting settings.

Secure cylinders upright. Keep oxygen equipment free of oil and grease. Check hoses, fittings, seats and tips for damage; use compatible equipment and the manufacturer-required reverse-flow and flashback protection. Confirm that the gas supply can sustain the selected tip flow. [10, 11]

Follow the site hot-work process: isolation and verification of the workpiece, fire control, ventilation and appropriate eye, face and body protection. Do not torch an unknown or closed vessel or a live line.

Use the exact equipment manual for connection, valve-opening and leak-test procedures. An approved leak solution is used, never a flame. Do not light equipment with a leak or damaged parts.

FIELD CHECK Acetylene delivery must not exceed 15 psig. That limit is not a recommended setting; the selected tip may require less. [10]

[10] OSHA 1910.253. [11] Miller setup guidance. See source links on page 50.

FIELD NOTES / TEST FEEDBACK

Gauge pressures by thickness

Example chart scope: Smith SC12 acetylene cutting tips on compatible SC-series equipment; Miller GE/27.0, June 2026, page 3. These are regulator delivery pressures in psig, not cylinder contents-gauge readings. [12]

Use only if the installed equipment matches. Other brands and even Smith MC12 tips have different values. Read the full current chart and equipment manual before setting the regulators.

Steel in	SC12 tip	Oxygen psig	Acetylene psig
1/4	SC12-0	30	10
3/8	SC12-0	35	10
1/2	SC12-1	35	10
5/8	SC12-1	40	10
3/4	SC12-2	36	10
1	SC12-2	41	10
1-1/2	SC12-3	42	10
2	SC12-3	47	10

FIELD CHECK Select by thickness being cut, not pipe diameter. A bevel has a longer cut path; use applicable manufacturer guidance, not an invented pressure increase.

[12] Selected factual values from Smith/Miller GE/27.0; original table layout. No universal pressure chart is implied.

Set, light and shut down

Delivery pressure is checked under the flow conditions specified for that torch, including cutting-oxygen flow when required. A closed-valve static reading may differ. Hose losses and flow restrictions matter. Purge oxygen and fuel separately in a ventilated area away from ignition sources. [11]

For the Smith cutting-attachment procedure in [13], after setup, leak testing and purging: open fuel about 1/8 turn and ignite with a friction spark lighter. Increase fuel until smoke clears; add preheat oxygen until the acetylene feather disappears into the inner cones. Follow the manual for stand-off and cutting technique.

For that Smith attachment, normal shutdown closes preheat oxygen, then fuel, then cylinder valves. Bleed fuel and oxygen separately as directed until their gauges show zero, close torch valves and back out regulator adjustments. Follow [13] for the full sequence; other equipment may differ.

Popping, sustained hissing, flame inside the torch or overheating requires stopping under the equipment emergency procedure and inspection before reuse. Do not keep cutting by turning pressures up.

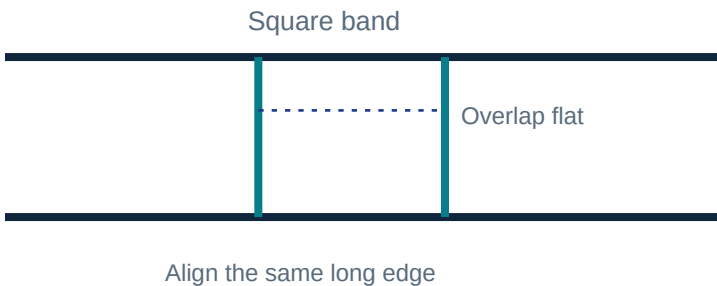
FIELD CHECK This is a companion to supervised training and the equipment manual. It is not a complete startup or emergency procedure.

[11] Miller setup guidance. [13] Smith OM-262659, sections 6 and 9; confirm the revision for your equipment.

FIELD NOTES / TEST FEEDBACK

Use a pipe wraparound

1. Secure the cold pipe against rolling. Clean the layout band. Select an undamaged wrap long enough to overlap and wide enough to stay square; obey its temperature limit. A wraparound is a marking guide, not a torch heat shield.
[14]
2. Mark the desired cut location from a verified datum. Bring one straight long edge of the wrap to that mark and wrap snugly around the pipe without twisting.
3. At the overlap, align the long edge with itself. Hold the layers flat. The edge must meet continuously with no step; a spiral or skewed overlap makes a sloping cut line.
4. Trace along that same edge around the full circumference, repositioning your hand without shifting the wrap. Check closure and measure from the datum at several positions. Remove the wrap before hot work.



FIELD CHECK A dented or out-of-round pipe can prevent a clean square wrap. Resolve the geometry rather than forcing a distorted guide.

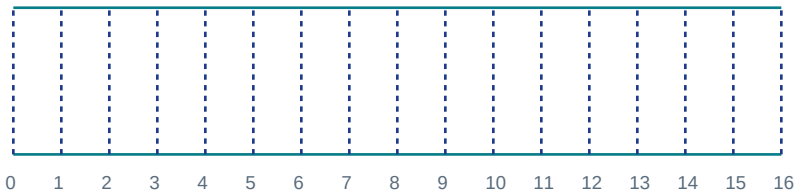
[14] Jackson Safety tool purpose; original marking workflow.

Cut and fold register paper

Use dry, unstretched register paper wide enough for the largest ordinate plus a holding margin. Keep it away from heat. If it is too narrow, use wider drafting paper; do not omit part of the curve.

1. Wrap the strip once around the actual branch OD with square overlapping edges. Mark where the starting end meets the strip. Unwrap and cut at that mark: the working length is exactly one circumference. Add any fastening tab later.
2. Bring the two short ends together and crease: two equal panels. Keep folding the stacked strip end-to-end three more times, aligning edges each time. Four halvings give 16 panels. Unfold completely.
3. Number the edges and creases 0 through 16. Draw each crease straight across the width, perpendicular to a common long-edge baseline. These are the ordinate station lines. Stations 0 and 16 become the same seam when wrapped.

16 panels = 16 equal arcs; seam is 0 / 16



Fold count: 1 → 2 panels; 2 → 4; 3 → 8; 4 → 16.

Each panel represents $360^\circ/16 = 22.5^\circ$.

To use 12 stations, measure $C/12$ instead; repeated halving does not produce 12 equal divisions.

FIELD CHECK Folding locates stations only. It does not determine the saddle heights. Calculate and mark those after unfolding.

Calculate a 90-degree saddle

Scope: round branch meeting a round main at 90°, with intersecting axes, branch radius r no greater than main radius R . Use actual outside radii. This is the outside-surface branch-end profile, not the opening in the main.

Look down the branch axis. Station 0 lies in the direction of the main axis; station 4 is 90° around the branch. At angle A , the side displacement is $y = r \times \sin(A)$. The main surface height is $z = \sqrt{R^2 - y^2}$.

The lowest contact height is $z_{min} = \sqrt{R^2 - r^2}$. Measure each cutback ordinate O from the plane of the longest fingers, toward the retained pipe: $O = \sqrt{R^2 - (r \times \sin(A))^2} - \sqrt{R^2 - r^2}$.

On paper, use the long edge as that longest-tip baseline. Plot O on each station into the retained side and connect with a fair curve. The region between baseline and curve is waste. Do not cut the stacked folded paper.

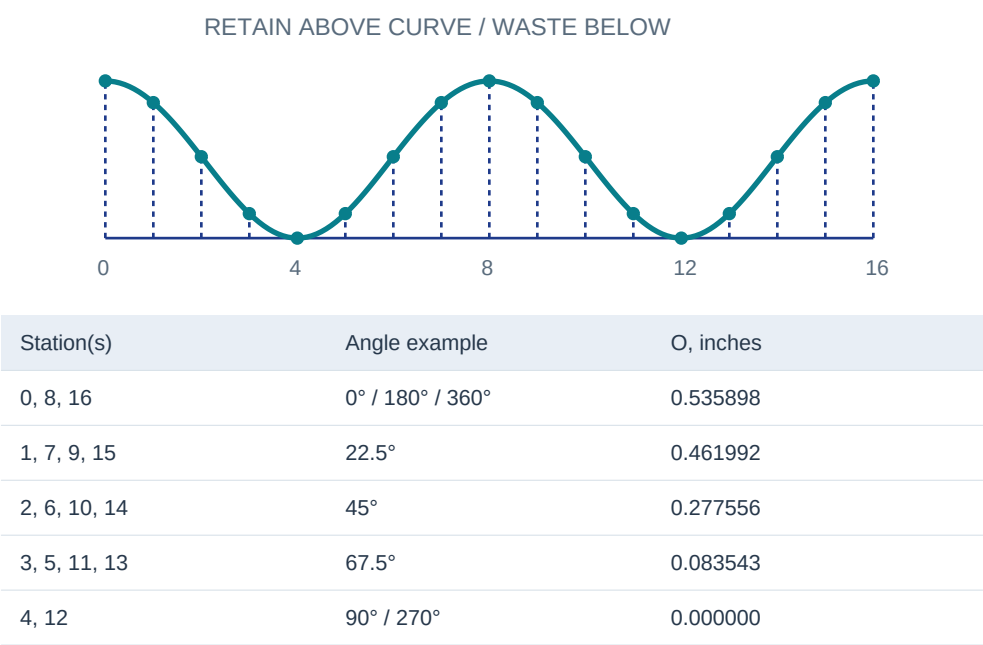
FIELD CHECK This formula excludes offsets, oblique branches, bevels, root gap and reinforcement. Pressure-piping branch design and weld preparation require the approved detail.

FIELD NOTES / TEST FEEDBACK

Worked saddle: 4 OD onto 8 OD

Illustrative actual ODs are 4.000 and 8.000 in, not nominal pipe sizes. $r = 2$ in; $R = 4$ in; $C = 12.56637$ in; station spacing = 0.785398 in.

At station 2, $A = 45^\circ$: $y^2 = (2 \times \sin 45^\circ)^2 = 2$. $O = \sqrt{16 - 2} - \sqrt{16 - 4} = 0.277556$ in.



FIELD CHECK All ordinates are nonnegative. The two longest fingers are at stations 4 and 12. The two deepest cutbacks are at 0 and 8. Diagram is schematic, not a full-size template.

Equal-size saddle and more stations

When $R = r$, $\text{sqrt}(R^2 - r^2) = 0$, so $O = R \times \text{abs}(\cos(A))$. The absolute value keeps the cutback nonnegative around the full circle.

For actual equal ODs of 4 in, $R = 2$ in. At 0° , 22.5° , 45° , 67.5° and 90° , the cutbacks are 2.0000, 1.8478, 1.4142, 0.7654 and 0.0000 in. Repeat by symmetry around the other quadrants.

Large pipes or rapidly changing curves may need more stations. Another halving produces 32 panels at 11.25° each. Recalculate ordinates at those angles; do not just average neighboring heights.

Equal-size 90° saddles have sharp changes at the longest fingers. Preserve those intersections in the geometric pattern; do not round them away by eye. Fit and finish still follow the approved joint detail.

At $A = 67.5^\circ$ and $R = 2$ in:
 $O = 2 \times \text{abs}(\cos 67.5^\circ) = 0.765367$ in.
For 32 stations, station 6 is $6 \times 11.25^\circ = 67.5^\circ$,
so it must reproduce that same ordinate.

FIELD CHECK The blank paper width must exceed the greatest ordinate plus a holding margin. A 2 in-wide strip cannot hold a 2 in ordinate and a useful margin.

FIELD NOTES / TEST FEEDBACK

Transfer, trim and check the saddle

- 1. With paper flat, connect calculated marks smoothly, preserving any genuine corner. Label main OD, branch OD, angle, station 0, retained side and baseline. Cut only the single-layer developed curve.
- 2. Wrap the paper around the branch, align stations 0 and 16 without adding overlap to the working circumference, and secure with a separate tape tab. Check that the baseline is square and the seam closes.
- 3. Align the station-0 direction with the main centerline plane. Trace the curve. Mark the scrap side clearly. Remove paper and wraparound before cutting.
- 4. Cut on the scrap side, leaving any finishing allowance required by the job. Trial-fit to a verified main or matching gauge and inspect the whole circumference. Apply the approved bevel and root gap separately.
- 5. Check axial length from the correct datum: this template references the longest fingers. A centerline-to-centerline dimension is not automatically the same length.

FIELD CHECK Practice on paper or scrap first. A close geometric fit does not qualify a pressure connection, establish reinforcement or approve a welding procedure.

FIELD NOTES / TEST FEEDBACK

Practical supplement: sources

Edition 0.2 adds formula practice, static rigging calculations, model-specific torch references and original paper-template instructions. Independently computed examples and diagrams are training material, not job approvals.

Saddle ordinates are derived here from the intersection of two perpendicular circular cylinders. Validate a paper trial on the actual diameters before fabrication. No commercial manual text or saddle table was copied.

A qualified rigging reviewer and oxy-fuel instructor must review these chapters before commercial release. Manufacturer values are selected factual data with attribution, not a licensed reproduction of a complete chart.

[7] Lift-All: Effect of sling angle

<https://lift-all.com/product-files/catalog/effect-of-angle-chart.pdf>

[8] OpenStax: Conditions for static equilibrium

<https://openstax.org/books/university-physics-volume-1/pages/12-1-conditions-for-static-equilibrium>

[9] OSHA 1926.251: Rigging equipment

<https://www.osha.gov/laws-regs/regulations/standardnumber/1926/1926.251>

Torch and marking references

Read the complete current equipment instructions. The retrieved Smith manual URL retains an older filename but serves revision H. Recheck the actual revision and chart before operation.

Settings on page 41 apply only to the named SC12 equipment family. To prepare another table, record the torch and attachment models, tip series, gas, thickness, hose arrangement and manufacturer source.

[10] OSHA 1910.253: Oxygen-fuel equipment

<https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.253>

[11] Miller: Oxy-fuel setup checks

<https://www.millerwelds.com/en-us/resources/knowledge-hub/welding-safety/10-steps-for-safe-oxy-fuel-torch-setup>

[12] Smith/Miller GE/27.0, June 2026, p.3

<https://www.millerwelds.com/-/media/miller-electric/import/specsheets/file/ge270-cutting-and-gouging-tips--english.pdf>

[13] Smith OM-262659, retrieved revision H

https://www.millerwelds.com/files/owners-manuals/O262659E_MIL.pdf

[14] Jackson Safety: Wrap-A-Round tools

<https://www.jacksonsafety.eu/product-page/wrap-a-round-tools-1>